Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Focused ultrasound technology for the direct degradation of polyfluoroalkyl substances (PFAS)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals			
	Medium Category	Small Category	Medium Category	Small Category
	Chemical Processes	*Process Equipment Technologies/ Process System Technologies*		
Concept of Proposed Technology	*The core concept of this technology can be summarized by these three functional pillars:* ○ Acoustic Cavitation & Energy Concentration: High-frequency sound waves (typically 400–600 kHz) are emitted from a transducer and focused into a central point in the water. This creates millions of microbubbles that expand and then collapse violently (implode), generating localized "hot spots" with temperatures reaching 5,000–10,000 K and pressures exceeding 500 bar. ○ Direct Pyrolytic Destruction: Because PFAS are surfactants, they naturally accumulate at the gas-liquid interface of these bubbles. When the bubbles collapse, the intense heat and pressure cause the pyrolysis (thermal decomposition) of the exceptionally strong carbon-fluorine (C–F) and carbon-carbon (C–C) bonds, breaking them			

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	down into fluoride ions, CO_2 and water. ㅇRadical-Induced Mineralization: The extreme conditions also dissociate water molecules into highly reactive hydroxyl ion and hydrogen radicals. These radicals provide an indirect degradation pathway that helps mineralize any remaining shorter-chain intermediates, ensuring higher removal efficiencies.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* This research aims to solve the PFAS wastewater problem—the final environmental puzzle in battery recycling—and secure technological leadership by shifting the waste disposal paradigm from incineration-centered approaches to direct submerged decomposition. ㅇOvercoming the Unique Challenges of Used Battery Recycling Processes and Limitations of Existing Technologies Addressing High-Concentration, Complex Pollutants: Wastewater generated during the leaching and solvent extraction processes of spent battery black mass contains not only high concentrations of PFAS but also heavy metals, organic solvents, and high-concentration salts. Existing activated carbon adsorption or ion exchange resins only separate PFAS without destroying it at the source, and their efficiency is severely impaired in high-concentration wastewater. ㅇEconomic Viability and Scalability of Modular Equipment Development for On-site Application On-site processing and flexibility: Battery recycling plants vary in scale and wastewater characteristics. Modular equipment can be immediately installed on the process line and flexibly scaled out in parallel according to wastewater generation, reducing initial capital expenditure (CAPEX) and operating expenses (OPEX). Secured a commercial process capable of stably reducing and rendering harmless pollutants to discharge standard levels through pilot-scale (≥1 ton/h) field demonstration.
Key Words	1. Lithium-ion battery recycling 2. High-concentration PFAS wastewater 3. Non-incineration PFAS treatment technology 4. Focused Ultrasonic Technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3rd Stage)	End of the Project	Level (6th Stage)
R&D Objectives	ㅇOptimization of Energy Efficiency and Treatment Throughput (Scale-out Strategy): Establish the engineering parameters for a scalable system where multiple ultrasound-hybrid modules can be connected in parallel or series. The goal is to optimize the Electric Energy per Order (EE/O)—the energy required to reduce PFAS concentration by one order of magnitude—ensuring the technology is economically competitive with traditional incineration by leveraging focused energy zones to minimize heat loss to the bulk fluid. ㅇ Pilot-Scale Validation in Real-World Battery Recycling Effluents: Transition from laboratory-scale tests to a containerized prototype for on-site validation at a battery recycling facility. This involves evaluating the system's ability to maintain high PFAS degradation rates in the presence of "interference" compounds like residual organic solvents, heavy metals (Ni, Co, Mn), and surfactants, ensuring that the modular unit can operate continuously with minimal maintenance and no hazardous secondary waste generation.			
R&D Target Specifications	ㅇ Development of a focused ultrasonic-based non-combustion hybrid decomposition technology and modular equipment targeting high-concentration PFAS-containing wastewater generated from wet recycling processes of spent batteries ㅇ Securing a commercial process capable of reliably reducing and rendering harmless wastewater to discharge standard levels through pilot-scale (≥1 ton/h) field demonstration			
R&D Plan	ㅇHigh-concentration fluoride non-combustion decomposition in lithium-ion battery (LIB) recycling process (wet method) ① Development of core process technology for high-concentration PFAS decomposition · Establishment of focused ultrasonic-based physical-chemical-catalytic hybrid decomposition mechanism · Derivation of optimal decomposition conditions: influence of output, frequency, residence time, pH, salinity, and metal ions ② Durable process design for high-salinity and acid/alkali conditions · Reflecting battery wet process characteristics: high salinity, acid/alkali (pH 2~12), heavy metals (Ni, Co, Mn, Li, etc.) · Design to prevent corrosion, scaling, and precipitation ③ Establishment of PFAS analysis, toxicity, and material balance evaluation system · Degradation performance evaluation through component analysis ④ Development of modular degradation equipment and process systems · Lab → Pilot → Field scale-up structure · Modular parallel/series design · Includes automated control, safety design, and long-term operational stability ⑤ Pilot Demonstration and Field Optimization · Field application of battery recycling wet process · ≥ 1 ton/h pilot demonstration · Operational logic calibration using demonstration data ⑦ Commercialization Package Development · Equipment design specifications, operating manuals Safety and equipment certification (KC, CE, etc.)			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	ㅇProvision of wastewater with various characteristics generated during the wet recycling process of spent batteries ㅇProcess development for creating non-incineration decomposition equipment suitable for the wet recycling process of spent batteries ㅇDemonstration verification and commissioning for reviewing equipment field application testing ㅇVerification of reduction rates and decomposition technology through PFAS analysis in wastewater during equipment operation
Expected R&R of Korean SME Partner	ㅇ Development of high-concentration fluorine non-combustion decomposition technology for lithium-ion battery (LIB) recycling processes (wet method) ㅇDevelopment of a continuous non-combustion decomposition system suitable for lithium-ion battery (LIB) recycling processes (wet method) ㅇDevelopment of PFAS decomposition technology and equipment capable of meeting European regulatory standards

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technical Achievement: Paradigm Shift in PFAS Treatment While conventional incineration methods consumed high energy and posed secondary risks of air pollution, this technology is innovative in adopting a non-thermal approach. • Secured high-efficiency decomposition mechanism: Utilized core technology that effectively breaks PFAS's strong carbon-fluorine (C-F) bonds via cavitation from focused ultrasound, enabling PFAS decomposition in secondary battery recycling processes. • Hybrid process optimization: Established a flexible treatment system combining ultrasound with other technologies (oxidants, adsorption, etc.) to handle variations in concentration and water quality. • Demonstration data acquisition: Proved commercially viable performance (meeting discharge standards) beyond the laboratory level through a pilot demonstration at a scale of ≥1 ton/hour. 2. Commercialization and Economic Value • Marketability of Modular Equipment: Developed as a 'modular' system, it can be immediately integrated as additional equipment into existing waste battery recycling plants, significantly reducing installation space and costs. • Operational Cost Reduction: Higher energy efficiency compared to high-temperature incineration lowers wastewater treatment costs, directly enhancing the price competitiveness of recycled battery materials • Global Regulatory Compliance: As PFAS discharge regulations tighten worldwide, this technology becomes an essential 'environmental compliance' solution for battery recyclers. 3. Industrial and Environmental Ripple Effects • Fulfilling the Circular Economy: By completely eliminating environmental pollutants generated during battery recycling, it completes a truly 'eco-friendly mobility ecosystem'. • Technology Domestication and Export: PFAS treatment technology is highly complex, potentially creating overseas dependency. Developing domestic equipment through this research enables preemptive capture of the global environmental market. • Cross-Sector Application: The technology can be horizontally deployed to other precision manufacturing industries, such as semiconductors and displays, where PFAS-containing wastewater is generated.

6. References

(1) Kim, S. Y., Shin, H., Lee, S., Hwangbo, M., & **Hwangbo, S.** (2025). Focused ultrasound technology for the direct degradation of per- and polyfluoroalkyl substances (PFAS): A sustainable approach to contaminant remediation. *Journal of Hazardous Materials Advances*, *15*, 100456. https://doi.org/10.1016/j.hazadv.2025.100456

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization